

AI × City Climate Action Hackathon 2026

Route2Zero

Transparent corridor intelligence for a just e-jeepney transition.

A transparent, export-ready decision-support MVP

Team Larpers · METRO MANILA · 2026

1,522

ROUTES SCREENED

GTFS route-direction records

71.4

TOP PRIORITY

Default scenario score

THE DECISION QUESTION

Which corridors deserve deeper electrification work first?

Route2Zero turns a fragmented evidence base into a shortlist that city teams can inspect, challenge and validate.

- **Screen 1,522 route-direction records consistently.**
 - Balance climate value, equity exposure and feasibility.
 - Export the same decision context into formal reports.
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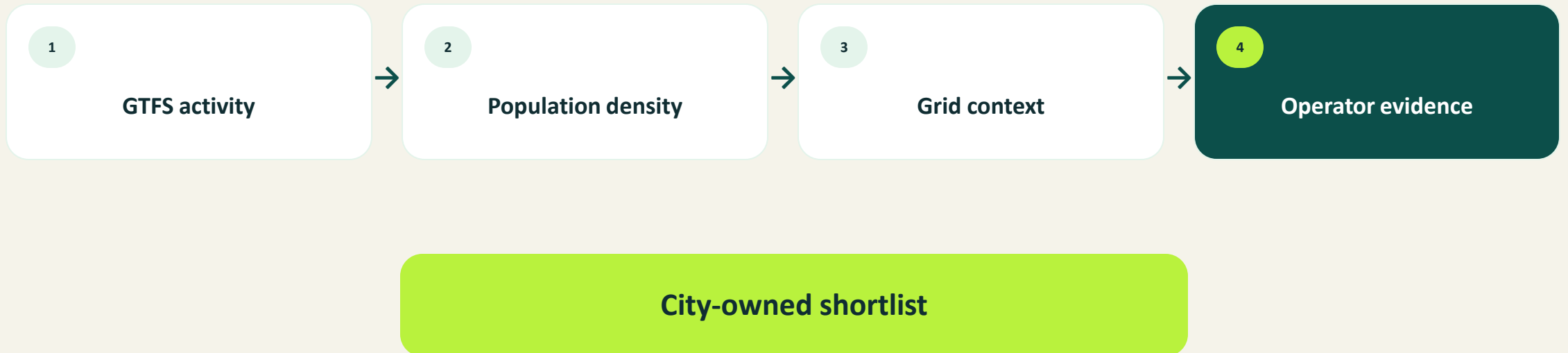
WHY THIS MATTERS

A route ranking without evidence can hard-code the wrong trade-offs.

- **High activity does not automatically mean high readiness.**
 - Equity proxies must be visible—not treated as ground truth.
 - Historic service data cannot stand in for present operations.
 - Operators and riders must be part of validation before investment.
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CURRENT PLANNING FRICTION

The evidence lives in different files, scales and institutions.



PRODUCT PROMISE

One transparent decision surface—without pretending uncertainty is precision.

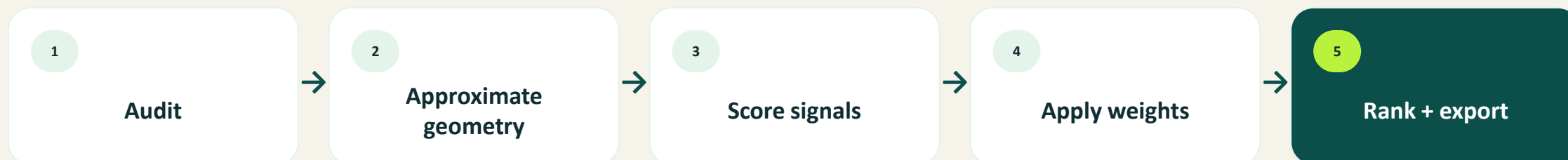
The rank is deterministic. Scenario controls are visible. AI can explain results, but it cannot change a score.

“

Transparent enough to challenge.
Useful enough to act on.

HOW THE PRODUCT WORKS

From source evidence to a challengeable action queue



DEFAULT POLICY LENS

Four interpretable signals, normalised to 100%

35%**Emissions potential**

Route activity proxy

35%**Equity density**

WorldPop catchment proxy

15%**Grid feasibility**

Luzon renewable-share proxy

15%**Operator readiness**

Neutral placeholder until verified

AI BOUNDARY

AI explains. The scoring pipeline ranks.

DETERMINISTIC

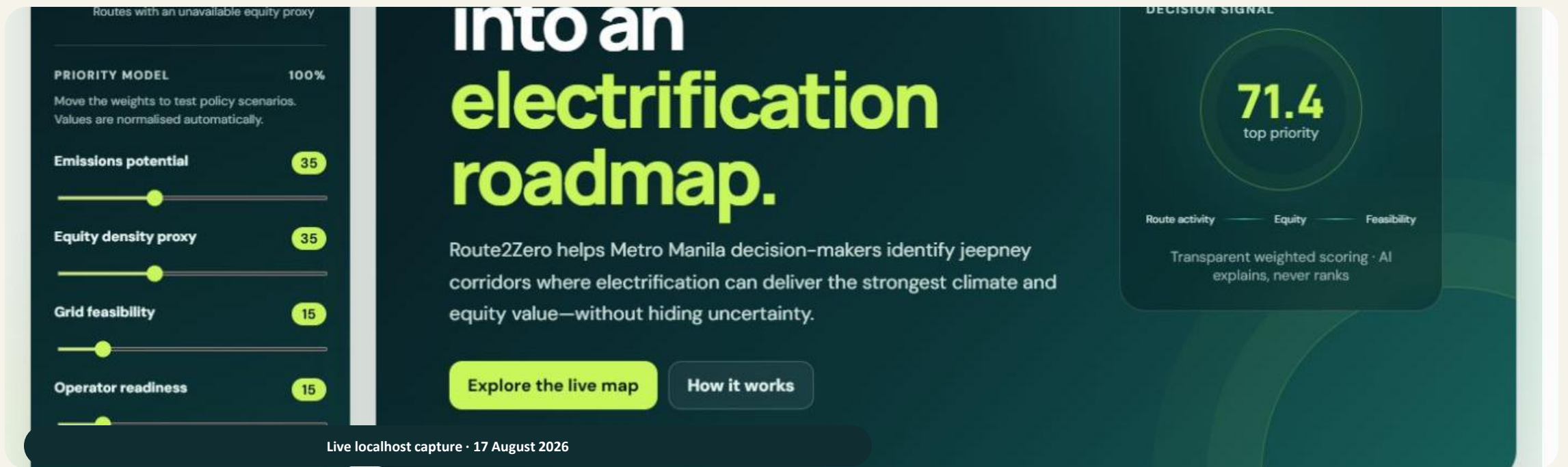
- Activity proxy
- Equity proxy
- Grid proxy
- Operator placeholder
- Weighted sum

OPTIONAL AI

- Plain-language rationale
- Evidence request summary
- Never edits score columns
- Never changes rank order

LIVE DASHBOARD

A 2026 decision cockpit built for judges and city teams



MAPBOX ROAD GEOMETRY

Selected routes follow the street network



SCENARIO TESTING

Weights update the action queue without hiding the policy choice.

- **Move emissions, equity, grid and operator sliders.**
 - Values are normalised automatically.
 - Every export records the current city lens and weights.
 - Reduced-confidence routes remain an explicit opt-in.
-

CURRENT ACTION QUEUE

Top routes for deeper feasibility work

Routes with an unavailable equity proxy

PRIORITY MODEL100%

Move the weights to test policy scenarios.
Values are normalised automatically.

Emissions potential35

Equity density proxy35

Grid feasibility15

Operator readiness15

#2	Dasmarinas Resettlement Area – Baclaran via Coastal Rd. LTFRB_PUJ1241	Pasay	69.7	97.5	73.3	16.1	50.0
#3	BACLARAN – DASMARINAS B BAYAN LTFRB_PUJ2083	Dasmarinas	69.5	96.7	73.6	16.1	50.0
#4	Binangonan – Pasig (TP) LTFRB_PUJ1156	Biñan	68.7	97.6	70.3	16.1	50.0
#5	Dasmarinas Resettlement Area – Baclaran via Coastal Rd. LTFRB_PUJ1240	Dasmarinas	68.2	93.8	72.6	16.1	50.0
#6	Francisco Homes – Cubao LTFRB_PUJ1352	Quezon City	68.1	89.7	76.6	16.1	50.0
#7	BACLARAN – DASMARINAS B BAYAN LTFRB_PUJ2084	Parañaque	67.7	91.4	73.6	16.1	50.0
#8	Binangonan – Pasig (TP)	Pasig	67.2	93.6	70.2	16.1	50.0

Top-ranked records under the default 35 / 35 / 15 / 15 policy lens.

LEADING SCREENING RESULT

Francisco Homes – Cubao

71.4

PRIORITY

Rank #1 under defaults

98.2

EMISSIONS

Activity proxy

77.4

EQUITY

Density proxy

16.1

GRID

Regional proxy

Operator readiness = 50.0 neutral placeholder

DECISION PACK

The dashboard exports the live policy context in three formats.

PDF

Print-ready decision report

DOC

Editable Word report

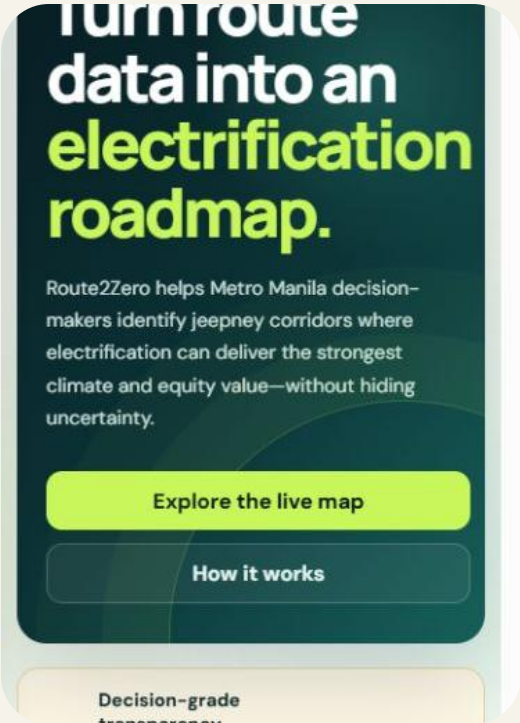
CSV

Scenario ranking data

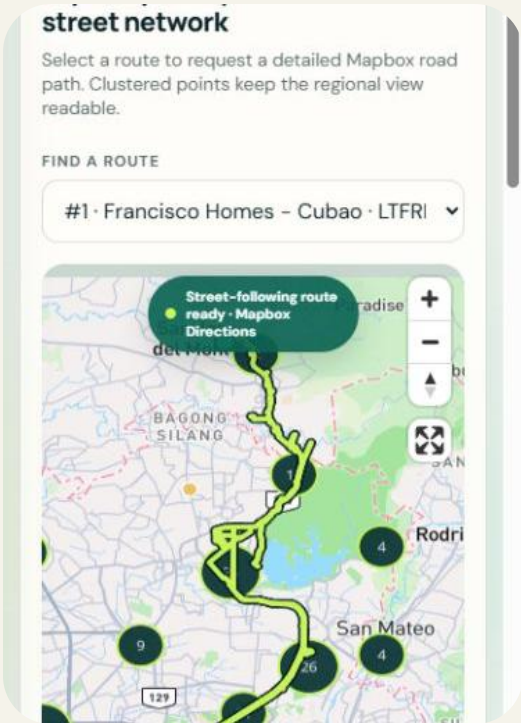
RESPONSIVE ACCESS

The same decision story works on mobile

390 × 844 viewport

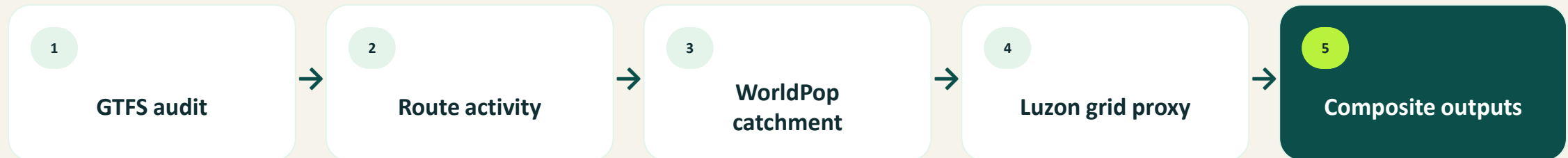


No horizontal overflow



REPRODUCIBLE PIPELINE

Evidence stays traceable from source fields to the interface.



WHAT THE MVP CAN PROVE

Strong on reproducibility; intentionally cautious on current operations.

Route universe	1,522 PUJ records	Reproducible
Score completeness	1,521 complete	Visible
Geometry	2 GTFS shapes; 1,520 approximated	Validate
Operator readiness	Neutral 50 placeholder	Acquire

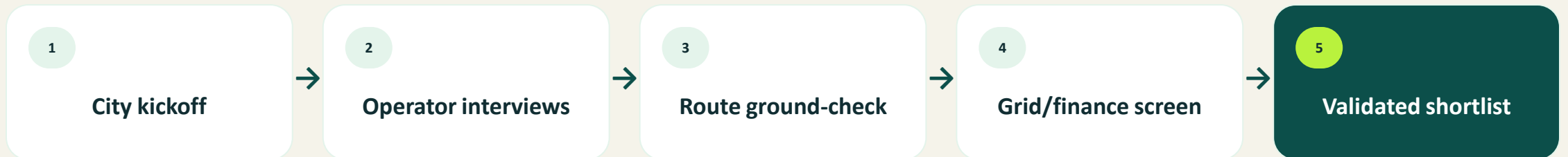
VISIBLE LIMITATIONS

Every important approximation becomes a validation task.

- **GTFS service calendar ends in 2020.**
 - 1,520 route records use stop-sequence geometry approximation.
 - WorldPop is density—not informal-settlement status.
 - Grid feasibility is regional, not depot-specific.
 - Operator readiness defaults to 50 until verified.
-

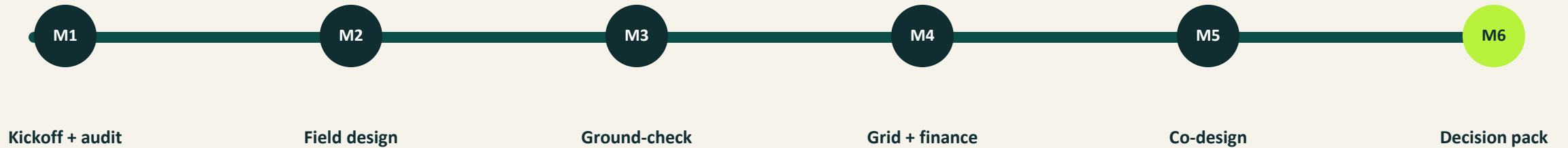
VALIDATION WORKFLOW

Convert uncertainty into an evidence-acquisition plan.



SIX-MONTH STRATEGY

Five workstreams, three milestone gates, one accountable decision pack.



MILESTONE GATES

Progress only when the evidence is decision-ready.

GATE 1

Evidence plan approved

End of M2

GATE 2

Routes validated

End of M4

GATE 3

City shortlist accepted

End of M6

ACCOUNTABILITY

A multidisciplinary team with named owners

John Marwin Ebona

Product lead

Prince Marl

Data lead

Joaquin Sarmiento

Mobility lead

Isaac Marcus

Engineering lead

Andrei Dela Cruz

Policy lead

Russel Mendez

Research lead

Tj Moreno

Engagement lead

JM Palaganas

QA + documentation

SUCCESS MEASURES

Judge the pilot by decision quality—not model novelty.

100%**shortlisted routes with evidence
cards**

Pilot target

3**LGU validation sessions**

Pilot target

≥80%**priority records ground-checked**

Pilot target

1**accepted decision pack**

Pilot target

THE ASK

Help Route2Zero turn a screening MVP into a city-owned electrification roadmap.

Pilot with three LGUs · validate with operators and riders · publish a transparent corridor shortlist.

SCREEN → VALIDATE → CO-DESIGN → ACT

route2zero · transparent by design